Current: charge flow rate

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate current as charge transferred per time.

SUCCESS CRITERIA - I can...

1. Use I=q/t with seconds and coulombs.
2. Describe conventional current direction.
3. Explain steady charge conservation at a junction.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “current” and “charge” to explain your model or result. Sentence starters are optional.

current: rate of charge flow

charge: a property responsible for electric interactions

interval: the time or range between two stated values

01 Warm-up: bring back an earlier idea

Both charge magnitudes become 3 times as large and separation becomes 5 times as large. Express new force as a fraction of the old.

Word help: separation = distance between the specified objects or centers

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $I = \Delta q/\Delta t$.
Current is charge per time; 1 A = 1 C/s. Conventional current follows positive charge motion.

Worked reasoning: If 12 C pass a point in 4 s, average current is I=Q/t=3 A. Charge circulates; a resistor does not consume charge.

Your turn: Draw a complete loop and label equal current before and after one resistor in steady state.

Sentence starter: The … in my model represents … . Its label shows … .



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Draw conventional current around the closed circuit, outside the battery from + toward -. What happens if a wire is disconnected?

Sentence starter: The … in my model represents … . Its label shows … .



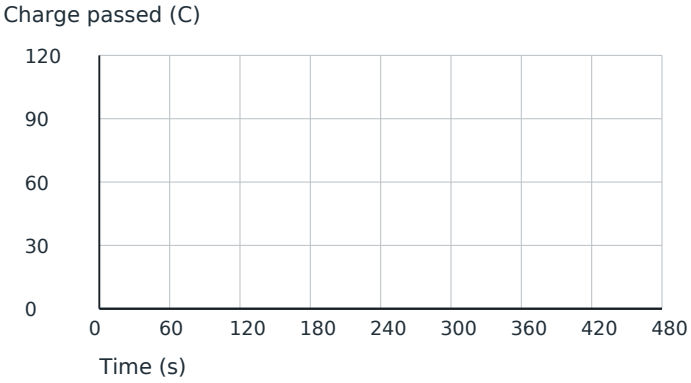
Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

04 Apply the physics

A wire carries a steady 0.25 A for 8 minutes.
Convert time to seconds, then calculate charge transferred.
Draw a cumulative-charge graph and label the slope.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

05 Explain your reasoning

At a junction, 5 A enters and 2 A leaves along one branch.
Find the other outgoing current in steady operation.
Explain the conservation rule.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A source approaching a stationary listener at 20 m/s emits 600 Hz.
Use $v_{sound}=340$ and $f_{obs}=f_s\ v/(v-v_s)$.
Find observed frequency.
Use wavefront spacing to explain the frequency change.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Source moves right. Add wavefronts.



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

Does a large electron drift speed have to equal the speed at which a switched circuit responds?
Explain qualitatively.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .



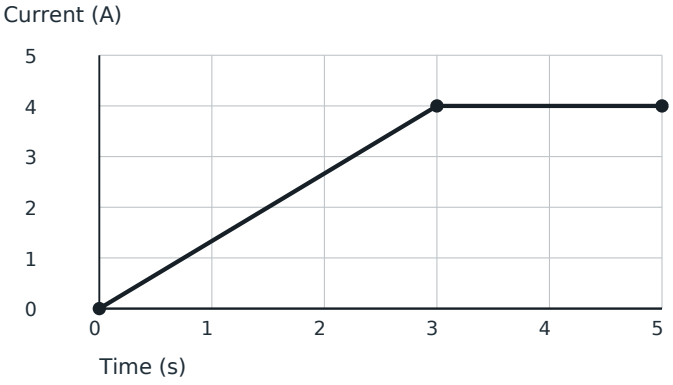
Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A current rises linearly from 0 to 4 A during 3 s, then stays 4 A for 2 s.
Find total charge from the current-time graph.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: 1 2 3